

1. Explain the various classes of data used for data processing with examples.

There are **3 types of data**:

1. Structured Data

- Data that has a **fixed and predefined structure**.
- Usually stored in a **database using rows and columns**.
- Easy to **store, search, update and process**.
- Uses a fixed schema.
- **Examples:** Student records, employee records, customer records.

Easy to remember:

👉 **Structured = Proper table**

Example:

ID	Name	Marks
101	Rahul	85
102	Priya	90

2. Semi-Structured Data

- Does **not have a fixed table structure**.
- But it still has **some organization or structure**.
- Usually uses **tags or metadata**.
- **Examples:** XML, JSON, HTML, emails.

Easy to remember:

👉 **Semi-structured = Some structure**

Example:

```
{  
  "name": "Rahul",  
  "marks": 85  
}
```

3. Unstructured Data

- Does **not have a predefined structure**.

- Cannot easily be stored in rows and columns.
- Difficult to organize and process using traditional databases.
- **Examples:** Images, videos, audio, presentations, letters.

Easy to remember:

👉 **Unstructured = No proper structure**

★ **Remember:**

Structured → **Table**

Semi-structured → **Tags/JSON/XML**

Unstructured → **Images/Video/Audio**

These three categories and their examples are directly covered in the PDF.

2. Discuss 3 V's of Big Data and Challenges of Big Data.

3 V's of Big Data

Big Data is mainly described using **3 V's**:

1. Volume → "How much data?"

- Means the **huge amount of data** generated and stored.
- Data comes from databases, sensors, social media, websites, videos, documents, etc.
- Because the amount is huge, we need systems capable of storing and processing it.

Remember:

👉 **Volume = Amount**

2. Velocity → "How fast?"

- Means the **speed at which data is generated, collected and processed.**
- Some data needs to be processed immediately.
- Examples:
 - Stock market data
 - IoT sensor data
 - Online transactions
 - Social media feeds

Processing can move from:

Batch → **Periodic** → **Near Real-time** → **Real-time**

Remember:

👉 **Velocity = Speed**

3. Variety → "What type?"

- Means the **different types and formats of data**.
- It includes:
 - Structured
 - Semi-structured
 - Unstructured

Remember:

👉 **Variety = Different types**

🧠 **Super easy memory trick:**

VOLUME = How much?

VELOCITY = How fast?

VARIETY = What type?

The PDF defines the three V's in exactly this way.

Challenges of Big Data

Remember these **6 challenges**:

1. **Exponential growth of data**
 - Data keeps increasing very quickly.
 - Difficult to identify useful information from unnecessary data.
2. **Cloud computing and virtualization**
 - Need suitable infrastructure to manage huge data.
 - Organizations must decide how and where to host Big Data systems.
3. **Data retention**
 - Difficult to decide **how long data should be stored**.
 - Some data remains useful for years, while some becomes useless quickly.
4. **Lack of skilled professionals**
 - Need people with skills in data science and Big Data technologies.
5. **Data processing challenges**
 - Difficulties in storing, searching, analyzing, transferring and securing huge amounts of data.
6. **Data visualization**
 - Huge amounts of data are difficult to present in a simple and understandable way.

Easy memory:

3. "It is easy to work with structured data compared to unstructured data." Justify with an example.

Basic idea

Structured data is easier because it is organized in a fixed format.

For example, consider a college database:

Student ID	Name	Course	Semester	Marks
S101	Rahul	MCA	2	82
S102	Priya	MCA	2	91

Everything is properly organized into **rows and columns**.

So we can easily:

1. Insert, Update and Delete

- Add a new student.
- Change a student's marks.
- Delete a student.

2. Security

- Different users can have different permissions.
- Example:
 - Student → Can view details.
 - Staff → Can modify marks.

3. Indexing

- Index helps us **find data faster**.
- Example: Create an index on the **Course** field to quickly find MCA students.

4. Scalability

- The database can grow from hundreds to thousands of students.
- More storage and processing capacity can be added.

5. ACID Transactions

Structured databases support **ACID** properties:

- **A – Atomicity:** All or nothing.
- **C – Consistency:** Data remains correct.
- **I – Isolation:** Transactions don't interfere with each other.
- **D – Durability:** Saved data remains saved even after a crash.

Why is unstructured data difficult?

Imagine you have:

-  Student photograph
-  Recorded lecture
-  Email
-  Voice recording

They don't have a fixed **row-column structure**.

Therefore, finding and processing specific information is more difficult.

★ Easy conclusion:

Structured data → **Organized** → **Easy to search/process**

Unstructured data → **Not organized** → **Difficult to search/process**

This question's example and the five advantages are given in pages 5–6 of your PDF.

4. Explain unstructured data. How to deal with it? Explain techniques used to find unstructured data.

What is Unstructured Data?

Unstructured data is data that:

- Does **not have a predefined structure**.
- Does not follow a fixed data model.
- Cannot easily be stored in rows and columns.
- Is difficult to process using traditional databases.

Examples:

- Images
- Videos

- Emails
- Audio
- Documents
- Social media content

Simple example:

A student table:

ID | Name | Marks

is structured.

But a student's:

 Video +  Photo +  Voice recording

is unstructured.

How do we deal with Unstructured Data?

The PDF gives these techniques:

1. Data Mining

Used to find **patterns and relationships** in large datasets.

Important techniques:

- **Association Rule Mining** → Finds what things occur together.
 - Example: People buying bread may also buy eggs.
- **Regression Analysis** → Predicts relationships between variables.
- **Collaborative Filtering** → Predicts what a user may prefer based on other users' preferences.

2. Text Analytics / Text Mining

- Extracts meaningful information from text.
- Finds patterns and trends.
- Includes:
 - Text categorization
 - Text clustering
 - Sentiment analysis
 - Entity extraction

3. Natural Language Processing (NLP)

- Helps computers **understand human language**.

Example:

"I really loved this movie!"

NLP can help a computer understand that the statement has a **positive sentiment**.

4. Noisy Text Analytics

Deals with messy text such as:

- Chats
- Blogs
- Emails
- Text messages

It handles things like:

- Spelling mistakes
- Abbreviations
- Missing punctuation
- Filler words such as "uh" and "um"

5. Manual Tagging with Metadata

- Manually add **metadata/tags** to unstructured data.
- This helps provide meaning and makes the data easier to understand.

6. POS Tagging

POS = **Part-of-Speech Tagging**

It identifies whether a word is:

- Noun
- Verb
- Adjective
- etc.

7. UIMA

Unstructured Information Management Architecture

- An open-source platform from IBM.
- Used for analyzing text and other unstructured data.
- Helps find hidden meaning and relationships.

 **Easy order to remember:**

Data Mining → Text Mining → NLP → Noisy Text → Metadata → POS → UIMA

5. Difference between Traditional BI and Big Data

This one is important because you asked about it earlier.

Traditional BI	Big Data
Data is stored in a central server	Data is stored in a distributed file system
Usually scales vertically — make the existing server more powerful	Scales horizontally — add more computers
Mainly analyzes data in offline mode	Analyzes data in real-time as well as offline mode
Mainly deals with structured data	Deals with structured + semi-structured + unstructured data
Data is moved to the processing function	Processing function is moved to the data

What does the last point mean?

Traditional BI:

DATA → Processing

You **bring the data to the processing system**.

Big Data:

DATA

↓

Processing is brought here

The **processing function goes to where the data is stored**.



Super-easy memory:

Remember these 5 differences:

Central → Distributed
Vertical → Horizontal
Offline → Real-time + Offline
Structured → All types
Data to Code → Code to Data

These five differences are directly from Question 5 on page 9 of your PDF.

Final 5-question revision

If you have very little time before the exam, remember this:

Q	Topic	What to remember
1	Types of Data	Structured, Semi-structured, Unstructured
2	3 V's + Challenges	Volume, Velocity, Variety + 6 challenges
3	Structured vs Unstructured	Structured is organized → easier to process
4	Unstructured Data	Definition + 7 techniques
5	Traditional BI vs Big Data	Central/Distributed, Vertical/Horizontal, Offline/Real-time, Structured/All, Data-to-Code/Code-to-Data

This way, you don't have to memorize long paragraphs. **Understand the bold idea first, then expand each point in the exam.**